

报关行BPMN

代码仓库

BPMN

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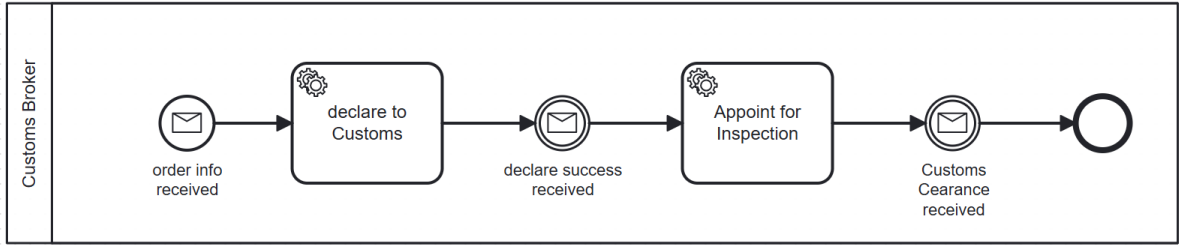
step2: 发送inspection-order

step3: 发送clearance-to-broker

代码仓库

<https://github.com/lxmm0427/customs-broker>

BPMN



业务理解

报关行收到订单信息后，依次完成报关申报、等待审批、预约检验、等待放行四个阶段，整个流程由外部系统通过消息驱动推进。

流程说明

步骤	元素类型	说明
order info received	Message Start Event	收到 <code>order-info-to-cb</code> 消息时自动创建新流程实例
declare to Customs	Service Task	触发 <code>declare-to-customs</code> Worker，生成报关单并向海关发布 <code>declaration-submitted</code> 消息
inspection order received	Intermediate Message Catch Event	挂起等待 <code>inspection-order</code> 消息（海关发出查验指令）

Appoint for Inspection	Service Task	触发 <code>appoint-for-inspection</code> Worker，预约查验并向海关发布 <code>inspection-appointment</code> 消息
Customs Clearance received	Intermediate Message Catch Event	挂起等待 <code>clearance-to-broker</code> 消息（海关发放放行证书）
order completed	End Event	流程正常结束

消息定义

入站消息（接收方：报关行）

消息名	发送方	关联键	说明
<code>order-info-to-cb</code>	货代	<code>orderId</code>	启动新流程实例，携带货物核心信息
<code>inspection-order</code>	海关	<code>orderId</code>	唤醒挂起实例，携带查验指令编号、类型和截止时间
<code>clearance-to-broker</code>	海关	<code>orderId</code>	唤醒挂起实例，携带放行证书信息

出站消息（发送方：报关行）

消息名	接收方	关联键	触发时机
<code>declaration-submitted</code>	海关	<code>orderId</code>	<code>declare-to-customs</code> Worker 执行时
<code>inspection-appointment</code>	海关	<code>orderId</code>	<code>appoint-for-inspection</code> Worker 执行时

Job Worker 说明

declare-to-customs

触发时机： 流程启动（收到 `order-info-to-cb` ）后立即触发。

职责： 生成报关单编号，向海关发布 `declaration-submitted` 消息，通知海关开始合规审核与风险评估。

读取的流程变量（来自 `order-info-to-cb` ，全部必填）

变量名	类型	说明
<code>orderId</code>	string	订单唯一标识（关联键）
<code>timestamp</code>	string	消息产生时间（ISO 8601）
<code>senderId</code>	string	货代系统标识，如 <code>FF-GLOBAL-LOGISTICS</code>
<code>cbId</code>	string	报关行标识，如 <code>CB-EXPRESS-02</code>
<code>hsCode</code>	string	货物海关编码，如 <code>85171210</code>
<code>cargoName</code>	string	货物名称，如 <code>Mobile Accessories</code>
<code>declaredValue</code>	number	申报价值
<code>currency</code>	string	货币代码（ISO 4217） ， 如 <code>USD</code>
<code>quantity</code>	number	申报数量
<code>countryOfOrigin</code>	string	原产国（ISO 3166-1 alpha-2） ， 如 <code>CN</code>
<code>countryOfDestination</code>	string	目的国（ISO 3166-1 alpha-2） ， 如 <code>US</code>

发布的出站消息： `declaration-submitted`

字段名	类型	必填	说明
-----	----	----	----

<code>orderId</code>	string	✓	订单唯一标识（关联键）
<code>timestamp</code>	string	✓	消息产生时间（ISO 8601），取申报时刻
<code>senderId</code>	string	✓	报关行标识，固定为 <code>SENDER_ID</code> （ <code>CUSTOMS-BROKER-05</code> ）
<code>declarationId</code>	string	✓	报关单编号，格式 <code>DECL-{orderId}-{timestamp}</code>
<code>hsCode</code>	string	✓	货物海关编码
<code>declaredValue</code>	number	✓	申报价值
<code>currency</code>	string	✓	货币代码（ISO 4217）
<code>quantity</code>	number	✓	申报数量
<code>countryOfOrigin</code>	string	✓	原产国（ISO 3166-1 alpha-2）
<code>countryOfDestination</code>	string	✓	目的国（ISO 3166-1 alpha-2）
<code>cargoDescription</code>	string	✓	货物描述（由流程变量 <code>cargoName</code> 映射而来）

appoint-for-inspection

触发时机：收到海关发出的 `inspection-order` 消息后触发。

职责：根据查验指令生成预约编号，向海关发布 `inspection-appointment` 消息，通知查验时间与地点。

读取的流程变量

变量名	必填	说明
-----	----	----

orderId	✓	订单唯一标识
declarationId	✓	报关单编号，由 declare-to-customs 生成
inspectionId	✓	查验指令编号，如 INSP-20260416-001
inspectionType	✓	查验类型： PHYSICAL （实货查验） / DOCUMENTARY （单证查验）
deadline	✓	完成查验的截止时间（ISO 8601）
reason	○	查验原因： RISK_ASSESSMENT / RANDOM

发布的出站消息：**inspection-appointment**

字段名	类型	必填	说明
orderId	string	✓	订单唯一标识（关联键）
timestamp	string	✓	消息产生时间（ISO 8601），取预约时刻
senderId	string	✓	报关行标识，固定为 SENDER_ID （ CUSTOMS-BROKER-05 ）
appointmentId	string	✓	预约编号，格式 APT-{inspectionId}-{timestamp}
appointmentTime	string	✓	预约查验时间（ISO 8601）
inspectionLocation	string	✓	查验地点

<code>contactPerson</code>	string	○	现场联系人（仅在流程变量存在时附加）
<code>contactPhone</code>	string	○	现场联系电话（仅在流程变量存在时附加）

流程变量

变量名	类型	来源	说明
<code>orderId</code>	string	<code>order-info-to-cb</code>	订单唯一标识，全流程关联键
<code>cbId</code>	string	<code>order-info-to-cb</code>	报关行标识
<code>hsCode</code>	string	<code>order-info-to-cb</code>	货物海关编码
<code>cargoName</code>	string	<code>order-info-to-cb</code>	货物名称
<code>declaredValue</code>	number	启动消息	申报价值
<code>currency</code>	string	启动消息	货币代码（ISO 4217）
<code>quantity</code>	number	启动消息	申报数量
<code>countryOfOrigin</code>	string	启动消息	原产国
<code>countryOfDestination</code>	string	启动消息	目的国
<code>declarationId</code>	string	<code>declare-to-customs</code>	报关单编号
<code>declaredAt</code>	string (ISO)	<code>declare-to-customs</code>	申报时间
<code>inspectionId</code>	string	<code>inspection-order</code>	查验指令编号

<code>inspectionType</code>	string	<code>inspection-order</code>	查验类型 (PHYSICAL / DOCUMENTARY)
<code>deadline</code>	string (ISO)	<code>inspection-order</code>	完成查验的截止时间
<code>reason</code>	string	<code>inspection-order</code>	查验原因, 可选 (RISK_ASSESSMENT / RANDOM)
<code>inspectionAppointmentId</code>	string	<code>appoint-for-inspection</code>	查验预约编号
<code>inspectionAppointedAt</code>	string (ISO)	<code>appoint-for-inspection</code>	预约时间
<code>inspectionLocation</code>	string	<code>appoint-for-inspection</code>	查验地点
<code>clearanceId</code>	string	<code>clearance-to-broker</code>	放行证书编号
<code>clearanceStatus</code>	string	<code>clearance-to-broker</code>	放行状态 (APPROVED / REJECTED)
<code>clearanceTime</code>	string (ISO)	<code>clearance-to-broker</code>	放行时间

运行测试

成功启动:


```
PS C:\Users\maomao\Downloads\customs-broker\customs-broker\nodejs> npm start

> customs-broker@1.0.0 start
> npx ts-node ./source/index.ts

Need to install the following packages:
ts-node@10.9.2
Ok to proceed? (y) y
Customs Broker workers started.
REST endpoint: http://localhost:8080
Workers registered:
  - declare-to-customs
  - appoint-for-inspection
Waiting for jobs...
```

step1: 发送 order-info-to-cb

The screenshot shows a REST client interface with the following details:

- Method:** POST
- URL:** http://localhost:8080/v2/messages/publication
- Body Type:** JSON
- Request Body:**

```
{
  "name": "order-info-to-cb",
  "correlationKey": "ORDER-20260416-001",
  "timeToLive": 10000,
  "variables": {
    "orderId": "ORDER-20260416-001",
    "timestamp": "2026-04-20T08:00:00Z",
    "senderId": "FF-GLOBAL-LOGISTICS",
    "cbId": "CB-EXPRESS-02",
    "bcId": "85171310"
  }
}
```
- Response:** 200 (227ms, 1.97K)
- Response Body:**

```
{
  "tenantId": "<default>",
  "messageKey": "2251799813703673"
}
```

Operate Dashboard Processes Decisions Operations Log

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Process Name: Customs Broker Process
Process Instance Key: 2251799813703674
Version: 1
Start Date: 2026-04-20 12:16:03
End Date: --
Parent Process Instance Key: None
Called Process Instances: None

order info received → declare to Customs → inspection order received → Appoint for Inspection → Customs Clearance received → order completed

Instance History

- Customs Broker Process
 - order info received
 - declare to Customs
 - inspection order received

Variables

Name	Value
cargoName	"Mobile Accessories"
cbId	"CB-EXPRESS-02"
countryOfDestination	"US"
countryOfOrigin	"CN"
countryOfTransit	"US"

step2: 发送 inspection-order

POST http://localhost:8080/v2/messages/publication

发送 保存 删除

Params Body 1 Headers Cookies Auth 前置操作 后置操作 1 设置

none form-data x-www-form-urlencoded JSON XML Text Binary GraphQL msgpack

自动生成 动态值 提取 格式化

```
1 {
2   "name": "inspection-order",
3   "correlationKey": "ORDER-20260416-001",
4   "variables": {
5     "orderId": "ORDER-20260416-001",
6     "inspectionId": "INSP-20260420-001",
7     "inspectionType": "PHYSICAL",
8     "deadline": "2026-04-21T10:00:00Z",
9     "reason": "RISK_ASSESSMENT"
10  }
11 }
```

Body Cookie Header 17 控制台 实际请求

200 · 24ms · 1.97K 校验响应 成功 (200) 通过

Pretty Raw Preview Visualize JSON utf8

```
1 {
2   "tenantId": "<default>",
3   "messageKey": "2251799813704336"
4 }
```

Operate Dashboard Processes Decisions Operations Log

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Process Name: Customs Broker Process Process Instance Key: 2251799813703674 Version: 1 Start Date: 2026-04-20 12:16:03 End Date: -- Parent Process Instance Key: None Called Process Instances: None

order info received → declare to Customs → inspection order received → Appoint for Inspection → Customs Clearance received → order completed

Instance History

- order info received
- declare to Customs
- inspection order received
- Appoint for Inspection
- Customs Clearance received

Variables

Name	Value
cargoName	"Mobile Accessories"
cbId	"CB-EXPRESS-02"
countryOfDestination	"US"
countryOfOrigin	"CN"
countryOfTransit	"US"

Add Variable + Copy variables

step3: 发送 clearance-to-broker

POST http://localhost:8080/v2/messages/publication

Body 1

JSON

```
{
  "name": "clearance-to-broker",
  "correlationKey": "ORDER-20260416-001",
  "variables": {
    "orderId": "ORDER-20260416-001",
    "clearanceId": "CLR-20260420-001",
    "clearanceStatus": "APPROVED",
    "clearanceTime": "2026-04-20T12:00:00Z"
  }
}
```

Body

```
{
  "tenantId": "<default>",
  "messageKey": "2251799813704582"
}
```

流程成功运行结束

Operate

DashboardProcessesDecisionsOperations Log

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Process Name

Customs Broker Process

Process Instance Key

2251799813703674

Version

1

Start Date

2026-04-20 12:16:03

End Date

2026-04-20 12:21:03

Parent Process Instance Key

None

Called Process Instances

None

Delete

order info received

declare to Customs

inspection order received

Appoint for Inspection

Customs Clearance received

1 completed

↺

+

-

↻

Instance History

End date

Execution count

Customs Broker Process

order info received

declare to Customs

inspection order received

Appoint for Inspection

Customs Clearance received

Variables

Listeners

Operations Log

Name

Value

cargoName

"Mobile Accessories"

cbId

"CB-EXPRESS-02"

clearanceId

"CLR-20260420-001"

clearanceStatus

"APPROVED"

clearanceTime

"2026-04-20T12:00:00Z"